Re-implementing and Stress-Testing POGS

A limitation-driven synthetic benchmark, an SE(3) Kalman motion prior,
and a real-time evaluation of Gaussian-splat object tracking

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Abstract

Persistent Object Gaussian Splat (POGS) tracks scanned objects by optimising their 6-DoF pose so that a Gaussian splat, rendered into the camera, matches the observed DINO features and stereo depth. The paper reports results on a physical robot at roughly 5 Hz and names its own limitations: objects must move slowly, and occlusion by hands or the gripper degrades tracking. This report describes (i) an audited re-implementation of the POGS pipeline that follows the paper and the authors' release (scan, train, group, track); (ii) a synthetic pick-and-place benchmark built around those limitations: 12 rendered conditions × 3 random trajectories, plus speed and depth-noise settings, giving 54 cases with exact ground truth; (iii) a constant-velocity SE(3) Kalman filter used as a warm start for the pose optimiser, with an optional loss check; and (iv) a real-time protocol in which a 30 Hz camera stream is tracked with the tracker's *measured* latency, so that a slow tracker skips frames and publishes stale poses. Offline, the Kalman variant with the loss check is the most accurate method overall (75% vs. 71% ADD-S success, 16% vs. 22% lost frames). It holds 100% at twice the object speed, where POGS falls to 82%, and it fixes the cereal-box failure (86% vs. 57%). In real time, POGS with its published 25 iterations runs at 1.9 Hz and is right on only 56% of camera frames. Cutting it to 10 iterations, or adding Kalman latency compensation, restores 100% at normal speed. No method survives a hand occluding the object, or the real-time stream played at twice the speed.

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1 Introduction

Pogs [1] represents a scanned scene as 3D Gaussians [2] that carry, in addition to colour, a distilled DINO feature [3] and a hash-grid field of CLIP [4] and instance-grouping features (in the style of LERF [5] and GARField [6]). After grouping the Gaussians into objects, each object's pose is tracked by gradient descent: the object's Gaussians are rendered under a candidate rigid transform and compared with the current camera frame's DINO features and stereo depth. The approach needs no CAD model and no re-scan, and the paper uses it for object reset and tool servoing on a physical robot.

The paper is candid about where this breaks: tracking runs at about 5 Hz, "objects had to be moved slowly", and occlusion (by the gripper or a hand) is a stated failure mode (Sec. VI of [1]). Physical experiments make these limits hard to quantify: there is no exact ground truth, and a condition cannot be varied while holding everything else fixed. This project therefore builds a controlled synthetic setting in which each limitation is a separate *factor* with several levels, re-implements Pogs faithfully, and tests a modification against it on exactly the same cases.

Contributions.
1. **v1: an audited Pogs re-implementation** (Sec. 4.1). Every default follows the paper or the release, and the source is cited in the code. Known deviations are listed explicitly (Table 4).
2. **A limitation-driven benchmark** (Sec. 3): one scanned floor scene with five household objects, and a robot gripper that picks, carries and places a target under controlled rotation, speed, occlusion, depth-noise, object and multi-object conditions.
3. **v2: an SE(3) Kalman motion prior** (Sec. 4.2), plugged into the unchanged tracking loop as a warm start, with velocity damping, speed limits and an optional loss check. A compute sweep at 25/10/5 optimiser iterations compares it with an equally cheap Pogs.
4. **v3: other targets and two simultaneous objects** (Sec. 4.3), drawn from the same run cache.
5. **A real-time protocol** (Sec. 4.4): 30 Hz renders of the same trajectories, a scheduler driven by measured latency, scoring at every camera frame, and Kalman latency compensation.

Grasp planning, tool servoing and the physical robot are out of scope.

2 Background: the POGS pipeline

Figure 1 summarises the four stages. We describe them as implemented in v1; Section 4.1 lists the hyperparameters.

Training. A splatfacto-style 3DGS model [8, 9] is fitted to the scan with an L1 + SSIM photometric loss and a Pearson depth loss. Each Gaussian also carries a 64-D feature, supervised from step 100 to match DINO ViT-S/14 features after a PCA from 384 to 64 dimensions. A 4-layer MLP lifts the rendered feature back. From step 2500, a multi-scale hash grid is supervised with CLIP embeddings (Huber loss) and with a contrastive instance-grouping loss on detector masks. The masks are Detic-like [10]; everything without a mask forms a "negative" mask. The PCA basis is part of the checkpoint: tracking frames must be projected with the *same* basis.

Grouping. Gaussians are clustered on their grouping features with HDBSCAN [11]. The release asks an operator to click the object to track. We select clusters geometrically instead, after the release's support-plane cut (Gaussians below table height + 1.5 cm are dropped). Clusters are rejected if they are a support surface (robust XY footprint), background (cosine similarity to the negative-mask target), or a fragment.

Tracking. The RigidGroupOptimizer [7, 1] keeps one rigid transform per object, a translation of the centroid and a world-frame rotation about it. At each frame it renders the object's Gaussians inside a region of interest (ROI) around the last estimate and minimises

$$\mathcal{L} \;=\; \frac{1}{|\Omega|}\sum_{u\in\Omega} \left\| \hat{f}(u) - f(u) \right\|_2 \;+\; \lambda_d \frac{1}{|\mathcal{V}|}\sum_{u\in\mathcal{V}} \big(\hat{d}(u) - d(u)\big)^2 \nVdash \Big[(\hat{d} - d)^2 < \tau_d^2\Big]\,, \tag{1}$$

where $\hat{f}, \hat{d}$ are the rendered DINO feature and depth, f, d the observed ones, $\lambda_d = 3.7$, and $\tau_d = 0.26\,\mathrm{m}$. Residuals above τ_d (occluders, background) are ignored. The optimiser runs Adam for 25 iterations per frame, with the learning rate decaying from 0.004 to 0.0008. The pose is initialised once, with 100 DINO-only iterations on the full frame followed by 70 iterations with depth and ROI.

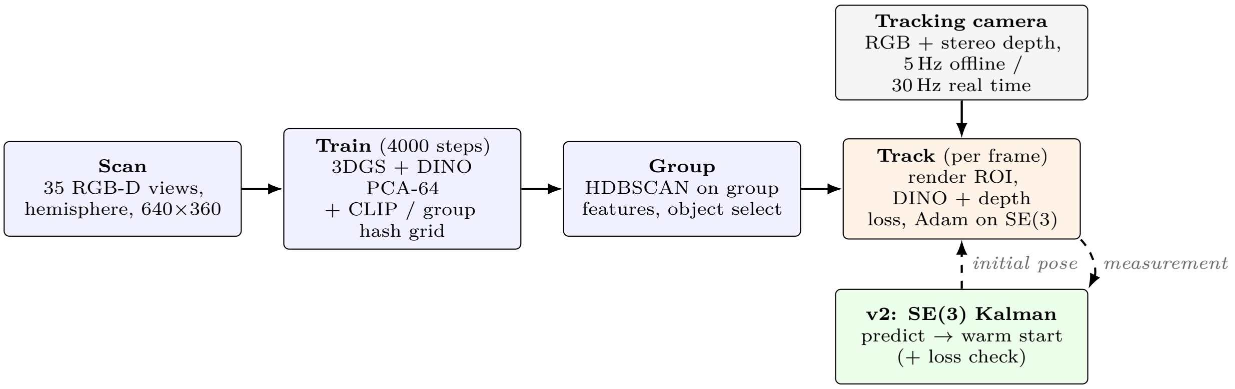

Tracking camera
RGB + stereo depth, 5 Hz offline / 30 Hz real time
Scan
35 RGB-D views, hemisphere, 640×360
Train (4000 steps)
3DGS + DINO PCA-64 + CLIP / group hash grid
Group
HDBSCAN on group features, object select
Track (per frame)
render ROI, DINO + depth loss, Adam on SE(3)
initial pose
measurement
v2: SE(3) Kalman
predict → warm start (+ loss check)

Figure 1: Pipeline. Blue: offline stages, run once per scene. Orange: per-frame tracking (v1). Green: the v2 motion prior, which only changes the optimiser's starting pose.

3 Synthetic dataset

All data come from an analytic ray-caster written for this project (`scene_generator/`). It renders RGB, metric depth and instance ids with shadows and a Blinn–Phong-style specular term. Because the generator knows every object's pose exactly, ground truth is exact. One scene and one trained model serve every tracking condition, so differences between conditions come from the condition alone.

3.1 Scene

The scene is a tiled floor with a low wooden staging table (40 × 32 cm, top at 6 cm) and five household objects standing on the floor around it (Table 1). The robot carries the target from the floor onto the table.

Table 1: Scene objects (scene v2). Size is the axis-aligned extent. All objects start on the floor; the table is the placement target.

Object	Size (cm)	Material (pattern, roughness)	Role	Target in
ceramic mug	10.4 × 7.6 × 9.7	glossy glaze (noise, 0.12), cream inside, coffee	default target	all factors
mustard bottle	7.2 × 4.8 × 16.3	plastic with a printed label band (stripes)	target	*object*
soup can	6.6 × 7.4 × 9.9	two-tone label (split), brushed tin lids	2nd mover	*multi-object*
red apple	7.6 × 7.6 × 8.2	skin patches (blotch)	distractor	–
cereal box	7.2 × 11.3 × 15.0	printed box art (blotch, blue/yellow)	target	*object*
table / floor	–	wood grain / tiles with grout	support, background	–

Scene v2. The first version of the generator (v1) evaluated procedural textures in *world* coordinates. A carried object therefore slid underneath a fixed pattern, which corrupts exactly the appearance features that POGS tracks. Version 2 evaluates every material in the hit primitive's own frame ($\mathbf{x}_{\text{loc}} = R^\top(\mathbf{x} - \mathbf{t})$), so textures move rigidly with the objects. Noise uses an integer lattice hash, so CPU and GPU renders are bit-identical. Version 2 also adds roughness and metallic parameters to the specular term. It fixes the table top at its nominal height (it was 6 mm higher, so placed objects sank into it) and moves the mug and soup can clear of the table footprint. A unit test checks that every footprint is at least 2 cm from the table and from every other object. Datasets, checkpoints and cached runs carry a `scene_version` and are regenerated when it changes. All results in this report use scene v2.

3.2 Scan (training data)

The scan has 35 views on a hemisphere of radius 0.68 m, at elevations 30–75° around the scene centre, rendered at 640 × 360 with a 62° field of view (Fig. 2). Depth gets stereo-like noise of 3 mm standard deviation at 0.7 m, growing linearly with range. The initial point cloud is back-projected from the depth maps and balanced per object (at most 2500 seed points per object, 4000 for the floor, 2500 for the table). Instance masks are filtered like Detic output: the floor and table are not reported, and masks larger than 1/3.5 of the image are dropped.

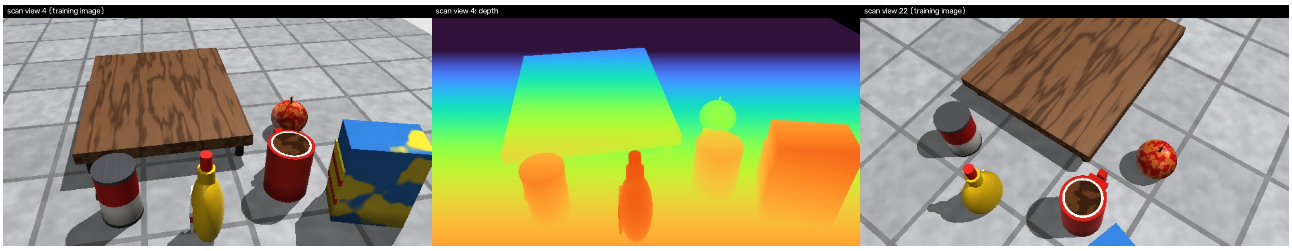


Figure 2: Two scan views (left, right) and the depth map of the left view (centre; near is warm). Materials: wood-grain table, tiled floor, glazed mug, labelled bottle and can, printed cereal box, patchy apple skin.

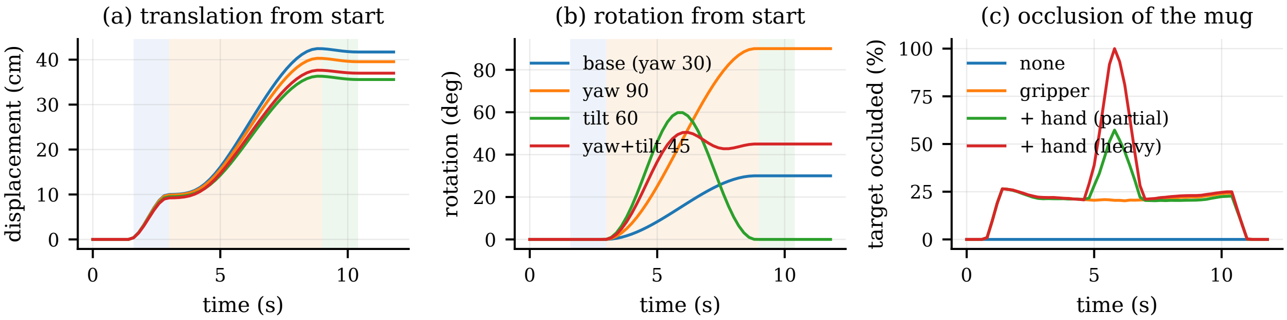


Figure 3: Ground-truth profiles (seed 0). (a) Translation of the target from its start; (b) total rotation angle; shading marks lift (blue), carry (orange) and place (green). (c) Fraction of the mug's pixels hidden by the gripper and hand: the gripper alone hides about 20% while it holds the mug; the hand sweep peaks at 57% (partial) and 100% (heavy).

3.3 Tracking sequences

A static tracking camera (889×500, $f_x = 700\,\mathrm{px}$, at $(0.24, -0.70, 0.54)\,\mathrm{m}$) watches a robot gripper (scaled to 0.7 of the stock model) pick up the target, carry it and place it on the table. The base rate is 60 frames at 5 Hz (12 s), the rate the paper reports. Each trajectory has five phases, defined as fractions of the object's active window:

0.00–0.12	approach	gripper descends onto the object (object static)
0.12–0.25	lift	to carry height (9–12 cm above the base, sampled)
0.25–0.75	carry	transit to the table with the condition's rotation profile
0.75–0.88	place	descend onto the table top (upright, final yaw)
0.88–1.00	release	gripper rises out of view (object static)

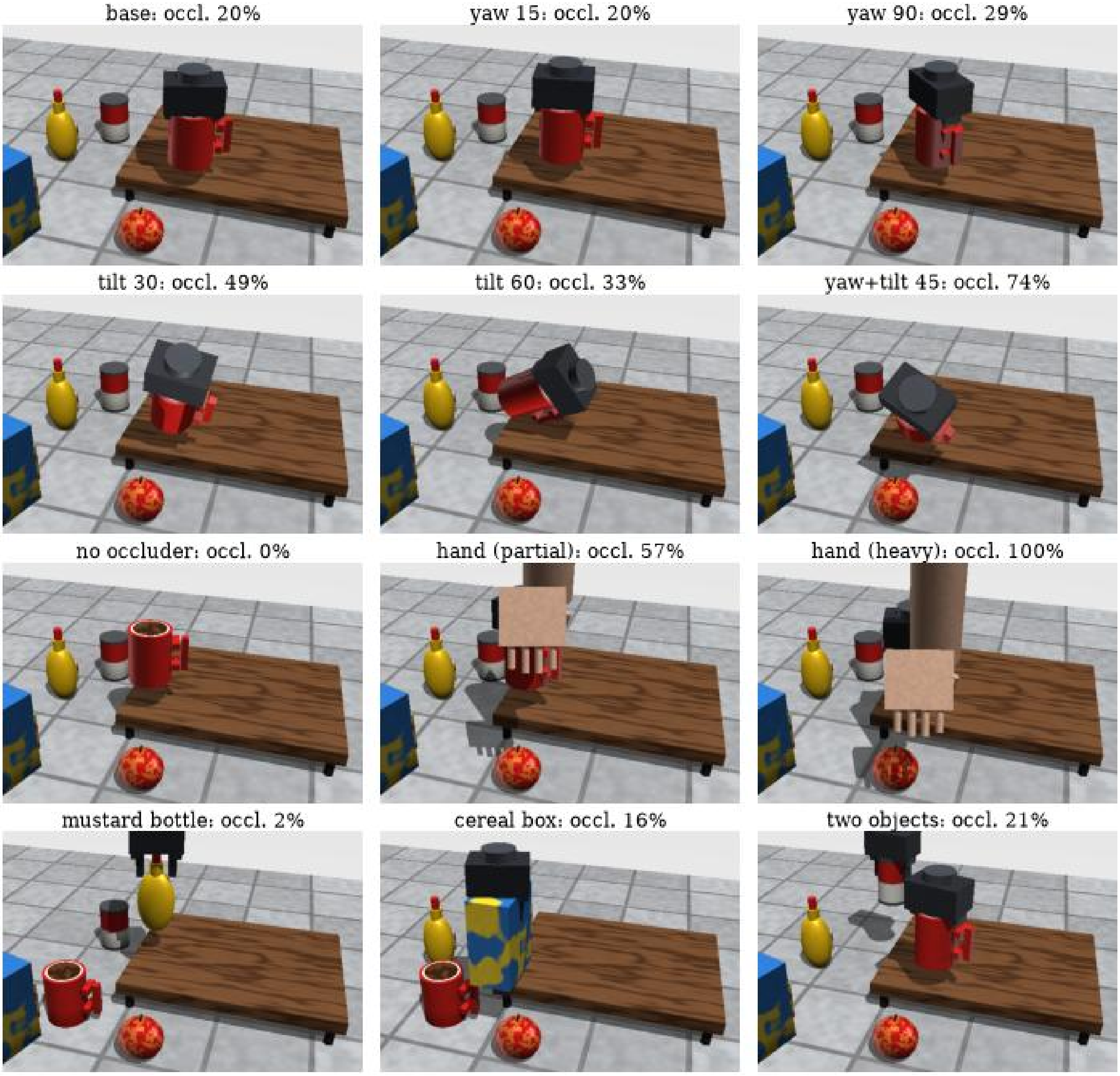
With $s \in [0, 1]$ the carry progress, the rotation profiles are: *yaw*, $\psi = \pm\theta s$; *tilt*, $\phi = \theta \sin(\pi s)$ about a random horizontal axis, upright again at placement; and *combined*, both together with the tilt peak equal to the total yaw. Each condition has three random trajectories (*seeds*). They vary the placement point on the table, the carry height, the path wobble, the rotation sign and the tilt axis. Figure 3 plots the resulting motion and occlusion profiles.

3.4 Conditions and factors

The base condition is: mug target, net yaw of 30°, gripper visible, 1× speed, 3 mm stereo depth noise. Each factor changes one aspect of the base condition (Table 2; one frame per rendered condition in Fig. 4). Speed and depth are applied at load time, without re-rendering. *Speed* k keeps every k-th frame at the same camera rate, so the object moves k times further per tracked frame; equivalently, the tracker is k times slower than the camera. *Depth* applies one of three noise profiles deterministically. With three seeds per level, the six factors expand to 54 tracking cases. The base case appears in every factor as the reference level.

3.5 Real-time sequences

For the real-time protocol (Sec. 4.4), the base trajectories are also rendered at 30 Hz: 355 frames, where frame $6k$ of the 30 Hz sequence is frame k of the 5 Hz one. The motion is identical; only the camera samples it six times more often. A playback speed of $k\times$ divides the timestamps by k without re-rendering.



base: occl. 20%
yaw 15: occl. 20%
yaw 90: occl. 29%
tilt 30: occl. 49%
tilt 60: occl. 33%
yaw+tilt 45: occl. 74%
no occluder: occl. 0%
hand (partial): occl. 57%
hand (heavy): occl. 100%
mustard bottle: occl. 2%
cereal box: occl. 16%
two objects: occl. 21%

Figure 4: One tracking-camera frame per rendered condition (seed 0, cropped): mid-carry for yaw conditions, peak tilt for tilt conditions, and peak occlusion for hand conditions. Titles give the fraction of the target hidden in that frame.

Table 2: Benchmark factors. Each level differs from the base case (**bold**) in one aspect only. Every level has three seeds.

Factor	Levels	Limitation probed (paper section)
rotation	yaw 15, **yaw 30**, yaw 90, tilt 30, tilt 60, yaw+tilt 45	large rotations; tilt exposes the unscanned underside (Sec. III assumptions)
speed	**1**×, 2×, 4× on base and on yaw 90	"objects had to be moved slowly" (Sec. VI)
occlusion	none, **gripper**, gripper + hand (partial, 57% peak), gripper + hand (heavy, 100% peak)	occlusion by gripper / human (Sec. VI)
depth	clean, **stereo 3 mm**, noisy 10 mm + 5% holes	reliance on stereo depth
object	**mug**, mustard bottle (thin, near-symmetric), cereal box (large, flat)	shape generality
multi-object	**one moving**, two moving (mug + soup can, two grippers)	simultaneous tracking

3.6 Ground truth and storage

Every frame stores RGB, depth in millimetres (`uint16`), instance masks, the phase, the rotation angle, per-object occlusion fractions, and the world-frame ground-truth motion of every object ($\mathbf{x}' = R\mathbf{x} + \mathbf{t}$, the same transform the renderer applies). Ground truth is used only for evaluation. It is never used to build groups, select objects or initialise poses, and a code rule enforces this.

4 Methods

4.1 v1: base POGS

v1 follows the paper and the release. Table 3 lists the settings that matter most. Each value lives in `common/config.py`, whose docstrings cite the release source.

Variants. The paper states that the tracking loss combines the feature and depth terms, but the release's per-frame `step()` defaults to `use_depth=False` and uses depth only during initialisation. v1 therefore reports both: **POGS** (paper, depth every frame) and **POGS (release)** (depth only at initialisation). The paper's ablations **No Depth** and **No DINO (RGB)** run on the base condition.

Protocol. As in the paper (Sec. IV-C), only the manipulated object(s) are tracked: the operator clicks the object to follow. Tracking all five objects drags the static ones whenever the gripper or the carried object passes through their ROI (12 cm drift in a smoke test).

4.2 v2: SE(3) Kalman motion prior

Motivation. POGS starts each frame's optimisation from the previous frame's estimate. When the object moves further between frames than 25 Adam steps can cover, the estimate lags, the ROI drifts off the object, and tracking is lost. A motion model can predict where the object will be and start the optimiser there.

State and dynamics. One filter runs per tracked object. Its state is expressed in exactly the parameters POGS optimises: the centroid translation $\mathbf{p}$ and the world-frame rotation $\mathbf{q}$, plus a linear velocity $\mathbf{v}$ and an angular velocity $\boldsymbol{\omega}$. Translation and rotation are two independent constant-velocity models driven by white-noise acceleration (Bar-Shalom et al. [12]). Rotation uses an error-state formulation: the nominal quaternion is propagated as $\mathbf{q} \leftarrow \exp(\boldsymbol{\omega}\,\Delta t) \otimes \mathbf{q}$, and the 3-D error lives in the tangent space, so there is no Euler singularity and no yaw-only restriction. For a step Δt and damping $\alpha = e^{-\Delta t/\tau}$,

$$F = \begin{bmatrix} I_3 & \Delta t\, I_3 \\ 0 & \alpha\, I_3 \end{bmatrix}, \qquad Q = \sigma_a^2 \begin{bmatrix} \frac{\Delta t^4}{4} I_3 & \frac{\Delta t^3}{2} I_3 \\ \frac{\Delta t^3}{2} I_3 & \Delta t^2 I_3 \end{bmatrix}, \qquad H = \begin{bmatrix} I_3 & 0 \end{bmatrix}, \tag{2}$$

with $\sigma_a = 0.5\,\mathrm{m/s^2}$ for translation and $4\,\mathrm{rad/s^2}$ for rotation. The measurement is the optimiser's converged pose, with $\sigma_p = 1\,\mathrm{cm}$ and $\sigma_\theta = 0.06\,\mathrm{rad}$.

Stabilisation. An undamped filter diverged by metres once tracking slipped. The warm-started optimiser's output is not independent of the prediction, so a wrong velocity re-confirms itself. v2 therefore damps velocity with $\tau = 1\,\mathrm{s}$ (velocity decays unless measurements sustain it), clamps speed at $|\mathbf{v}| \leq 0.5\,\mathrm{m/s}$ and $|\boldsymbol{\omega}| \leq 3\,\mathrm{rad/s}$, and records the normalised innovation squared (NIS) for consistency checks.

Table 3: Key v1 settings (all as in the paper or the release).

Stage	Setting
Training	4000 steps; DINO from step 100, CLIP + grouping from step 2500; splatfacto learning rates and densification (every 100 steps, gradient threshold 0.0008); SH degree 3; L1+SSIM ($\lambda = 0.2$) + 0.05×Pearson depth; gradient accumulation 3 for the hash-grid field
DINO	ViT-S/14, long side 1260 px, PCA 384→64 stored in the checkpoint, 4-layer MLP
Grouping	HDBSCAN (min. cluster 300, min. samples 50), 3-NN feature smoothing; support cut at 1.5 cm; support surface if 5–95% XY footprint $> 0.06\,\mathrm{m}^2$; background if similarity to the negative target > 0.98
Tracking	DINO L2 + 3.7×depth (ignore 0.26 m); Adam, lr 0.004 → 0.0008, 25 iterations per frame; ROI inflation 25% + 75 px, ROI $\leq$ 490 px; match resolution 500 px; only the selected objects are rendered
Initialisation	100 DINO-only iterations on the full frame + 70 with depth and ROI; lr 0.005; from the scan pose

Table 4: Intentional, documented deviations from the release.

Deviation	Reason
Geometric object selection	replaces the operator's click; same support-plane cut as the release
Simulated Detic masks	instance masks from the renderer, filtered like Detic (floor and table join the negative mask)
CPU HDBSCAN on a subsample	cuML not available; at most 25k points clustered, labels propagated
No camera-pose optimiser	synthetic camera poses are exact
Mask fallback	if the accumulation > 0.9 mask is empty, fall back to half the peak instead of crashing (counted)
Faster group transform	compose the K group transforms, then `index_select`; same maths as the Warp kernel, about 8× faster

Variants. **POGS + KF** starts every frame from the prediction. **POGS + KF + check** renders the tracking loss at both the prediction and the previous estimate before optimising, and keeps the prediction only where it lowers the loss (greedy, per object). This guards against bad predictions. A *fuse* variant, which replaced the optimiser's output with the filtered estimate, double-counted the prior and was dropped. A speed claim needs an equally cheap baseline, so v2 also runs every method at 10 and 5 optimiser iterations per frame, next to plain Pogs at the same budget.

Latency compensation. The filter accepts a variable Δt, so it can predict across the uneven gaps of the real-time protocol. It can also extrapolate an estimate to "now" without changing its state. With damping, the expected displacement over a horizon Δ is the damped velocity integral

$$\mathbf{p}(t+\Delta) = \mathbf{p}(t) + \mathbf{v}\,\tau\,(1 - e^{-\Delta/\tau}), \qquad \mathbf{q}(t+\Delta) = \exp\big(\boldsymbol{\omega}\,\tau(1 - e^{-\Delta/\tau})\big) \otimes \mathbf{q}(t). \tag{3}$$

Methods marked **+ latency comp.** publish this extrapolated pose. The optimisation itself is unchanged, so they share the run of the base method.

4.3 v3: other targets and two moving objects

v3 uses the same scene, checkpoint and grouping as v1 and v2. It reports the *object* factor (mug, mustard bottle, cereal box) and the *multi-object* factor (mug and soup can carried at the same time by two grippers) for Pogs, Pogs (release) and Pogs + KF. All three share the run cache, so cases that v1 or v2 already tracked are not repeated.

4.4 Real-time protocol

The offline benchmark tracks *every* frame and ignores compute time. A deployed tracker sees a camera stream instead. While it works on one frame, new frames arrive, and the pose it publishes describes the past. The real-time protocol (Algorithm 1) simulates this with the tracker's *measured* compute time on this machine.

Besides accuracy, the protocol reports the latency (compute per tracked frame), the tracker rate, the number of frames skipped per tracked frame, and the *estimate age*: how old the camera frame behind the published pose is. Timings depend on the GPU and on anything else running on it, so all real-time runs were made with the GPU otherwise idle (one NVIDIA RTX 4070).

Algorithm 1 Real-time evaluation (`common.evaluate.track_realtime`)

```
Require: camera timestamps $t_0..t_{N-1}$ (30 Hz, divided by the playback speed), tracker, optional Kalman prior
initialise on frame 0 (untimed: the object is at rest); $t_{\text{free}} \leftarrow 0$, $\ell \leftarrow -1$
while true do
    $j \leftarrow$ newest frame with $t_j \leq t_{\text{free}}$ and $j > \ell$; if none, $j \leftarrow \ell + 1$    ▷ skip stale frames
    if $j \geq N$ then break
    end if
    $s \leftarrow \max(t_{\text{free}}, t_j)$; predict the KF over $t_j - t_\ell$ (true gap)
    track frame $j$; measure compute time $c$ (GPU-synchronised)
    publish the estimate of frame $j$ at time $s + c$; $t_{\text{free}} \leftarrow s + c$; $\ell \leftarrow j$
end while
score every camera frame $i$ on the last estimate published at or before $t_i$
        (+ latency comp.: that estimate extrapolated by Eq. (3) to $t_i$)
```

5 Evaluation protocol

Metrics. Errors are computed against the exact ground truth of the target object, at every frame.

- **ADD-S success**: the fraction of frames whose ADD-S (the mean distance from each transformed model point to its closest ground-truth point, which is symmetry-aware [13]) is below 10% of the object diameter. This is the main metric.
- **Lost**: the fraction of frames with centroid position error above 10 cm.
- **Median position error** (cm) and **mean rotation error** (deg).
- The paper's **3 cm & 5°** rule. We report it but do not rank methods by it. Whenever the gripper is in view, it stays at 10–20% for every tracker (about 35% without an occluder), because the gripper tilts the held mug by ∼20° in the estimate (Sec. 6.4).

Baselines and statistics. Every table includes **Static (no motion)**, which always predicts the scan pose. It is right while the object rests (≈20% of frames) and wrong afterwards, so it is the floor that a tracker must beat. Means are over three seeds, and ± is the sample standard deviation across seeds. Methods are compared on identical cases only. A method name is a key of a shared run cache (`data_results/benchmark_runs`), so one name always means one configuration and no run is repeated across v1, v2 and v3.

6 Results

6.1 Reconstruction and grouping

The model trains in 11.7 min to 332k Gaussians, with a train-view PSNR of 34.1 dB and a mean depth error of 0.63 cm (Fig. 5). Scene v1 reached 31.3 dB and 2.6 cm. Grouping finds six HDBSCAN clusters. Selection keeps the five objects and rejects the floor and table (Fig. 6), with IoU against the true object Gaussians of 0.98 (mustard), 0.95 (apple), 0.88 (cereal), 0.80 (mug) and 0.67 (soup can). Cluster purity is 0.87, and there are no missing or duplicate objects.

On scene v2, the release thresholds first rejected three objects. A few hundred stray floor Gaussians in the mustard and soup clusters inflated their min/max footprint past the support-surface limit. The mug's grouping feature also had a similarity of 0.94 to the background target, above the threshold of 0.8, even though true background sits at 0.999. A 5–95th-percentile footprint and a 0.98 threshold fix both; no ground truth is involved in the selection.

6.2 v1: base POGS per factor

Figure 7 and Table 5 give the per-factor results at the published budget of 25 iterations per frame. On the base case, POGS tracks the mug on 100% of frames, with a median error of 0.78 cm. The benchmark separates the conditions clearly:

- **Speed** is the sharpest limit: 100% at 1×, 82% ± 31 at 2× (one seed lost), and 40% at 4×, with 60% of frames lost. Static gets 20%.
- **Large rotations** degrade tracking with high seed variance: yaw 90 reaches 49%, tilt 30–60 and yaw+tilt 45 reach 59–66%.
- **A hand occluding the object** breaks tracking: 43% (partial) and 34% (heavy), with ∼53% of frames lost. The gripper alone does not.

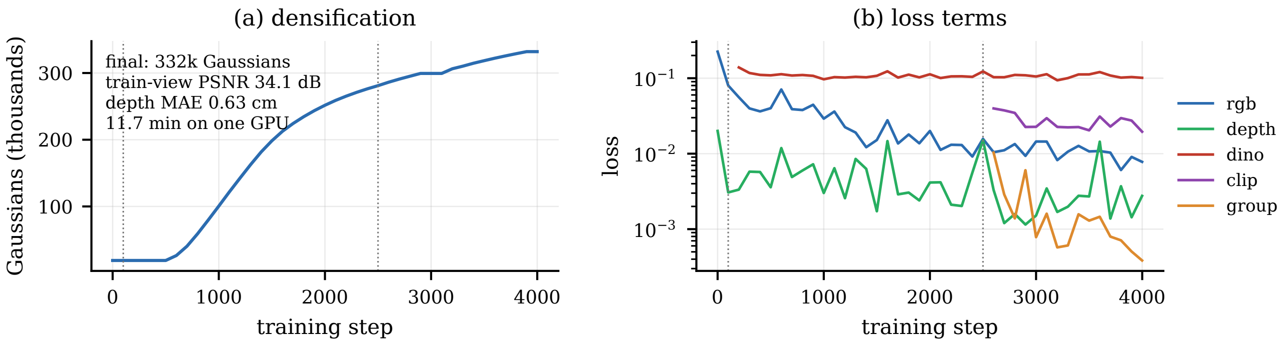


Figure 5: Training. (a) Number of Gaussians (densification); (b) loss terms on a log scale. Dotted lines mark the start of DINO (step 100) and CLIP + grouping supervision (step 2500).

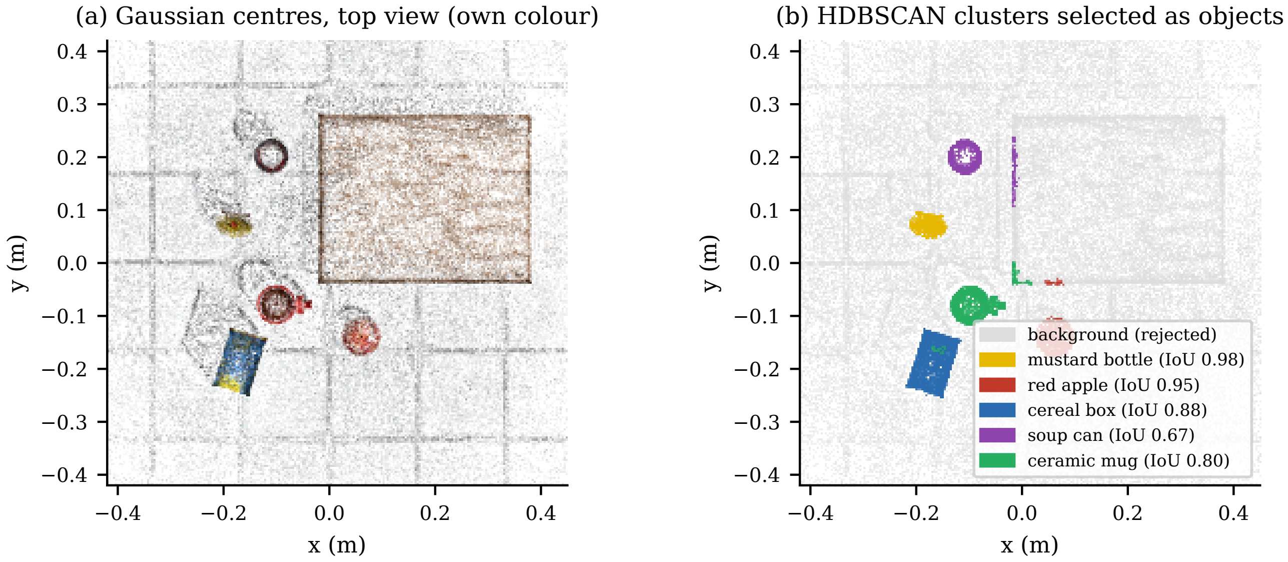


Figure 6: Grouping, top view of the Gaussian centres. (a) Each Gaussian in its own colour. (b) The five clusters selected as objects; floor and table are rejected. The IoU against the true Gaussians of each object is in the legend.

- **The cereal box** fails at 57% ± 13: once the estimate lags, the rendered box overlaps floor pixels, and the depth term pulls it into the background (Fig. 11).
- **Not discriminative** here: depth noise up to 10 mm + 5% holes (the 26 cm ignore threshold absorbs it), the mustard bottle, and two objects moving at once. All are at ≈100%.

6.3 Paper ablations

Table 6 shows that per-frame depth is essential. Without it (No Depth, and the release's depth-only-at-initialisation variant), success drops to 44–47% and the median error rises from 0.78 cm to more than 6 cm. DINO matters less than in the paper: plain RGB also reaches 100%, with a slightly higher median error (1.09 vs. 0.78 cm). The synthetic materials are textured but simpler than real objects. On scene v1, which had flatter colours, the two were indistinguishable.

6.4 Gripper-induced rotation error

The rotation error is ∼23° even when position tracking is perfect. Figure 8 shows the cause. Without an occluder, the mug's rotation error stays near 9°, mostly yaw lag: the mug is nearly a cylinder, and only the handle constrains yaw. With the gripper, the error rises to ∼25° while the mug is held and recovers after release. The gripper's depth lies in front of the mug, within the 26 cm ignore threshold, and the depth term tilts the estimate towards it. This is the occlusion limitation of the paper in a mild form. It also explains why the 3 cm & 5° rule saturates for every method.

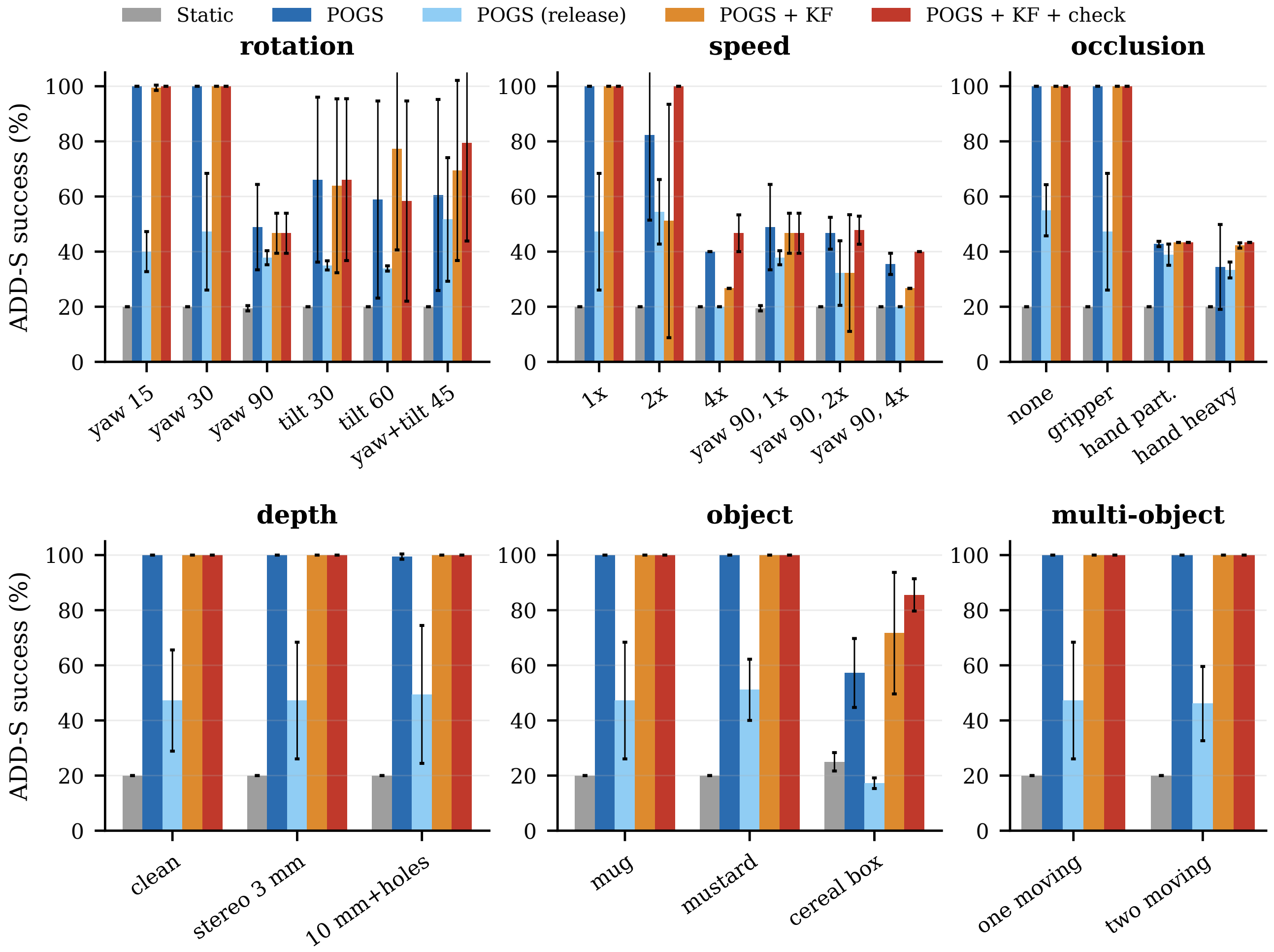


Figure 7: ADD-S success per factor level (25 iterations per frame; mean over three seeds, error bars = seed std). Static is the no-tracking floor.

6.5 v2: Kalman motion prior

Per factor. At 25 iterations (Table 5), the loss-checked filter is never meaningfully worse than POGS on success, and it clearly helps in three places:

- 2× speed: 100% on every seed, against 82% ± 31.
- 4× speed: 47% with 36% lost, against 40% with 60% lost.
- The cereal box: 86% ± 6 with no lost frames, against 57% ± 13 with 31% lost.

On large rotations, the KF variants roughly halve the lost-frame rate (yaw 90: 10% vs. 21%; yaw+tilt 45: 7% vs. 23%), but success changes within the seed spread. Hand occlusion is unchanged: no variant reaches the object while it is hidden. The plain KF without the check is unreliable. At 2× it reaches 51% ± 42 (one seed perfect, the others lost), and it is worse than POGS on several levels.

Compute budget. Figure 9 and Table 7 summarise all 54 cases. At 25 and at 10 iterations, the loss-checked filter has the highest success and the lowest lost rate. The extra loss evaluation costs 6–20% more time per frame. At 10 iterations (3.6 Hz) it matches POGS at 25 iterations (70% vs. 71%) at twice the frame rate, though it loses more frames (24% vs. 22%). At 5 iterations the prior hurts: plain KF collapses to 31%, and even with the check it reaches only 44%, below POGS at 5 iterations (55%). A weak optimiser cannot correct a wrong warm start, and the error feeds back into the velocity. The KF variants also have higher mean rotation error (34–52° vs. 28–36°), because the prior extrapolates any spurious angular velocity.

Speed vs. budget. Figure 10 crosses the speed factor with the budget. The benefit of the prior grows with object speed only when the optimiser has enough iterations to refine the warm start. At 25 iterations, the check variant is flat at 100% up to 2×. At 10 iterations it still improves 2× (66% vs. 50%). At 5 iterations it is worse than POGS even at 1×.

Case studies. Figure 11 shows per-frame position errors for four seed-0 cases.

Table 5: Per-factor results at 25 iterations: ADD-S success ± seed std / lost-frame rate. Best tracker per row in bold where methods differ.

Factor	Level	Static	POGS	POGS (release)	POGS + KF	POGS + KF + check
rotation	yaw 15	20 / 73	100±0 / 0	40±7 / 11	99±1 / 0	100±0 / 0
	yaw 30	20 / 75	100±0 / 0	47±21 / 23	100±0 / 0	100±0 / 0
	yaw 90	19 / 75	**49±15** / 21	38±3 / 43	47±7 / **10**	47±7 / **10**
	tilt 30	20 / 74	**66±30** / 6	35±2 / 42	64±32 / 6	**66±29** / 7
	tilt 60	20 / 67	59±36 / 37	34±1 / 39	**77±37** / **18**	58±36 / 22
	yaw+tilt 45	20 / 73	61±35 / 23	52±22 / 36	69±33 / **6**	**79±36** / 7
speed	1×	20 / 75	100±0 / 0	47±21 / 23	100±0 / 0	100±0 / 0
	2×	20 / 73	82±31 / 17	54±12 / 16	51±42 / 4	**100±0** / **0**
	4×	20 / 73	40±0 / 60	20±0 / 56	27±0 / 62	**47±7** / **36**
	yaw 90, 1×	19 / 75	**49±15** / 21	38±3 / 43	47±7 / **10**	47±7 / **10**
	yaw 90, 2×	20 / 74	47±6 / 37	32±12 / 51	32±21 / **19**	**48±5** / 46
	yaw 90, 4×	20 / 71	36±4 / 60	20±0 / 56	27±0 / 62	**40±0** / **53**
occlusion	none	20 / 76	100±0 / 0	55±9 / 0	100±0 / 0	100±0 / 0
	gripper	20 / 75	100±0 / 0	47±21 / 23	100±0 / 0	100±0 / 0
	+ hand, partial	20 / 76	43±1 / 54	39±4 / 54	43±0 / 54	43±0 / 54
	+ hand, heavy	20 / 70	34±15 / 52	33±3 / 54	42±1 / 55	**43±0** / 53
depth	clean	20 / 75	100±0 / 0	47±18 / 33	100±0 / 0	100±0 / 0
	stereo 3 mm	20 / 75	100±0 / 0	47±21 / 23	100±0 / 0	100±0 / 0
	10 mm + holes	20 / 75	99±1 / 0	49±25 / 34	100±0 / 0	100±0 / 0
object	mug	20 / 75	100±0 / 0	47±21 / 23	100±0 / 0	100±0 / 0
	mustard bottle	20 / 73	100±0 / 0	51±11 / 5	100±0 / 0	100±0 / 0
	cereal box	25 / 70	57±13 / 31	17±2 / 59	72±22 / 14	**86±6** / **0**
multi-object	one moving	20 / 75	100±0 / 0	47±21 / 23	100±0 / 0	100±0 / 0
	two moving	20 / 69	100±0 / 0	46±13 / 33	100±0 / 0	100±0 / 0

Table 6: Paper ablations on the base condition (3 seeds, 25 iterations).

Method	ADD-S success	Lost	Median pos. (cm)	Rot. err. (deg)
Static (no motion)	20%	75%	21.99	15.0
POGS (DINO + depth every frame)	**100%**	**0%**	**0.78**	23.1
No DINO (RGB + depth)	**100%**	**0%**	1.09	22.7
No Depth (DINO only)	44%	17%	6.46	25.4
POGS (release: depth only at init)	47%	23%	6.14	26.7

- 2× speed: POGS falls behind during the carry and never recovers, while both KF variants stay below 2 cm.
- Cereal box: POGS and KF + check track it, but the plain KF diverges near placement.
- Tilt 60: the plain KF follows the object, while POGS and KF + check lose it at peak tilt, when the unscanned underside faces the camera.
- Heavy hand occlusion: every method loses the mug when the hand covers it, and the plain KF then extrapolates it away at the speed limit.

These cases show why the plain KF is high-variance: the same mechanism that rescues a lagging estimate also amplifies a wrong one.

6.6 v3: other objects and two moving objects

The mustard bottle is tracked at 100% by every depth-using method, with a lower rotation error than the mug (8° vs. 23°): it is tall, and the gripper hides less of it. The cereal box is the hardest object (Table 5). Its large flat faces sit close to the floor, so a lagging estimate overlaps background depth. The loss check is what makes the KF safe here: 86% vs. 72% for the plain KF. With two objects moving at once (mug and soup can, each tracked with its own transform), all depth-using methods stay at 100%. The cost per frame roughly doubles (about 1.0 s for POGS), because each object has its own ROI render. The KF variants have a lower median error (0.60–0.70 cm vs. 0.96 cm).

6.7 Real-time tracking

Table 8 and Figures 12–14 give the real-time results on the base trajectory at 1× and 2× playback.

Published POGS cannot keep up. At 526 ms per frame, POGS processes one of every ∼15 camera frames (Fig. 13a). The pose it publishes is on average 0.8 s old and up to 1.2 s old (Fig. 13b). The same trajectory that

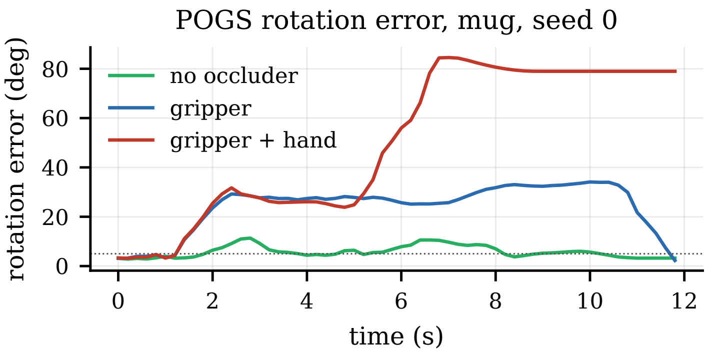


Figure 8: Pogs rotation error on the mug (seed 0) without an occluder, with the gripper, and with gripper + hand. The dotted line marks the paper's 5° threshold.

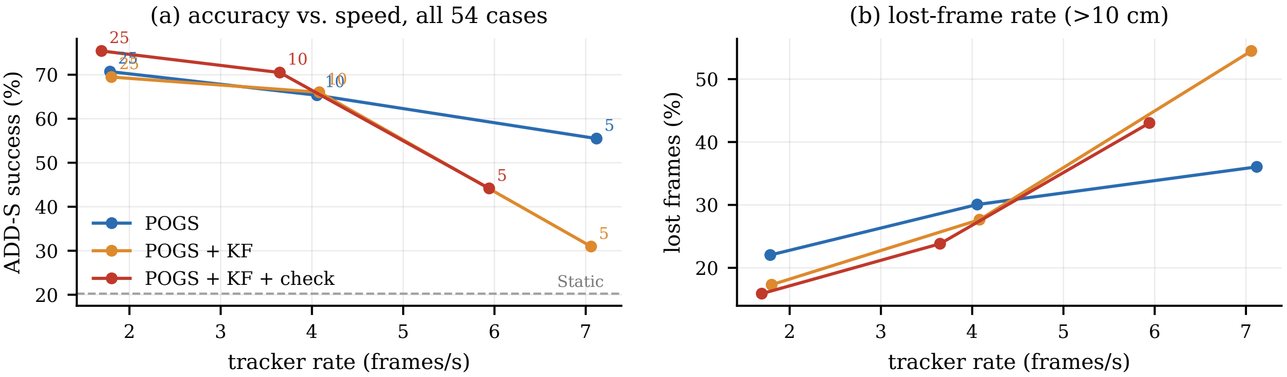


Figure 9: Accuracy vs. tracker rate over all 54 offline cases. Labels give the optimiser iterations per frame. (a) ADD-S success; (b) lost-frame rate.

it tracks perfectly offline now yields 56% ± 19 success, and on one seed the mug is lost for good (Fig. 14, left). Speed is the main fix: Pogs at 10 iterations runs at 4.4 Hz and is right on 100% of frames.

Latency compensation. Extrapolating the published estimate to the current time (Eq. 3) is free at run time and helps most when the tracker is slow. At 25 iterations it raises KF + check from 66% to 91%. At 10 iterations it reaches 100% with the lowest median error of all methods (1.49 cm vs. 1.69 cm for Pogs at 10 iterations). The saw-tooth error in Fig. 14 (left) is the estimate ageing between updates; compensation flattens it.

2× playback breaks everything. With the stream played twice as fast, every method falls to 20–34%. The best is KF + check + latency compensation at 25 iterations (34% ± 17, 29% lost), but with high variance across seeds. The motion prior does not rescue the 10-iteration tracker here.

Visual output. Every run can produce a dashboard video (Fig. 15). It shows the camera image with ground-truth (green) and estimated (cyan) boxes, the object splats rendered at the estimate over the image, the raw Gaussian scene with the tracked object moved to the estimate, and the raw object splats alone. Live text gives the latency, the tracker rate, the frame behind the shown estimate, its age, and the frames skipped.

Table 7: Compute budget, all 54 offline cases (rate = mean frames per second during tracking on an RTX 4070).

Method	Iters	Rate (Hz)	ADD-S success	Lost	Median pos. (cm)	Rot. err. (deg)
POGS	25	1.8	71%	22%	8.6	35.6
POGS + KF	25	1.8	69%	17%	9.9	41.7
POGS + KF + check	25	1.7	**75%**	**16%**	**6.4**	40.1
POGS	10	4.1	65%	30%	9.5	28.5
POGS + KF	10	4.1	66%	28%	11.4	41.1
POGS + KF + check	10	3.6	**70%**	**24%**	**9.4**	33.9
POGS	5	7.1	**55%**	**36%**	**11.0**	28.0
POGS + KF	5	7.1	31%	54%	29.4	51.6
POGS + KF + check	5	5.9	44%	43%	15.5	38.1

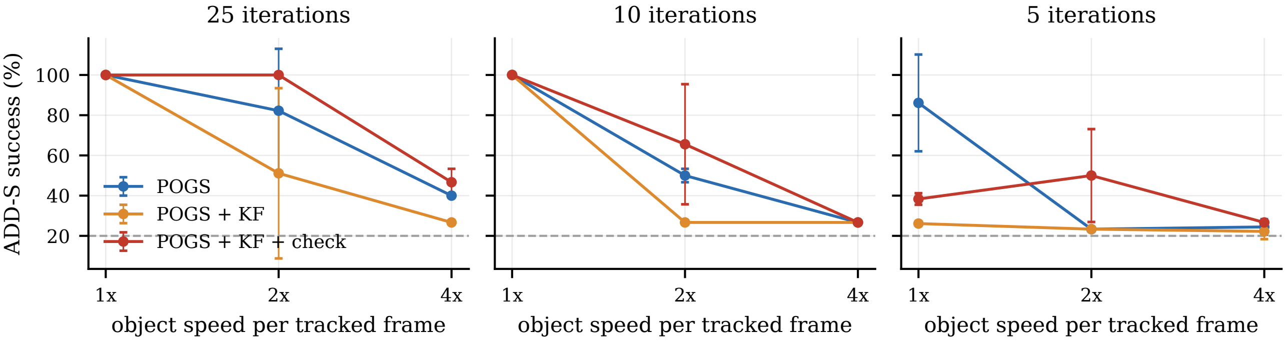


Figure 10: ADD-S success vs. object speed per tracked frame (base trajectory, mean ± seed std) at three optimiser budgets. Dashed: Static.

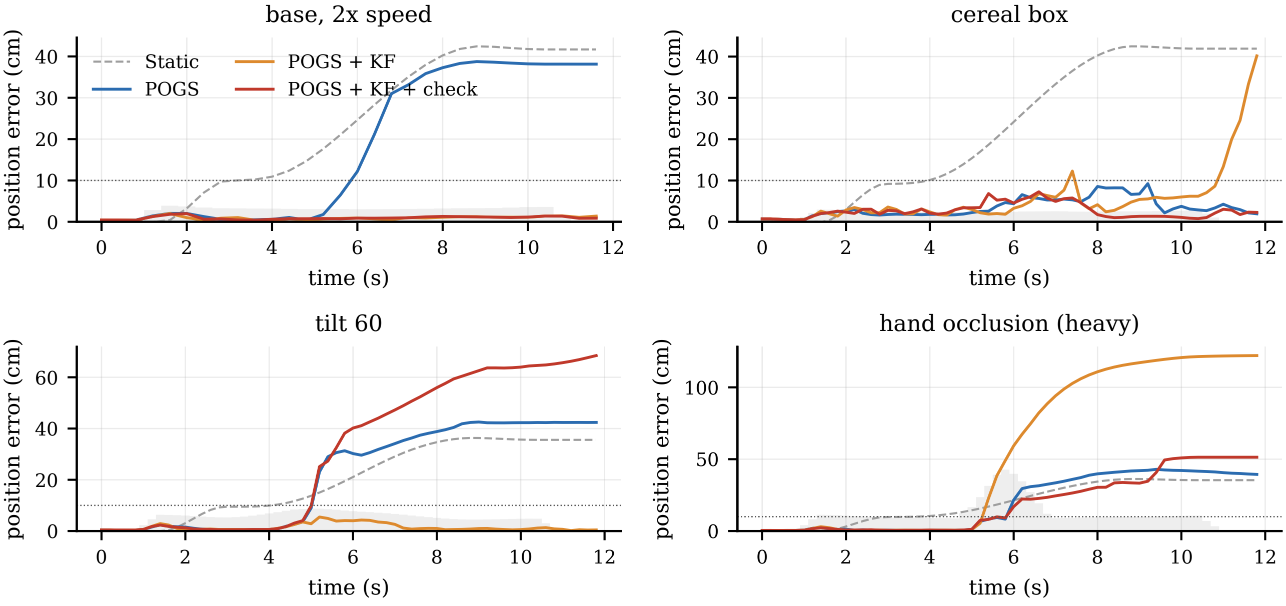


Figure 11: Per-frame position error, seed 0. Grey shading shows how much of the target is occluded; the dotted line marks the 10 cm “lost” threshold.

Table 8: Real-time results (30 Hz camera, base trajectory, 3 seeds). "Now" scores the published pose at every camera frame. Latency is the measured compute per tracked frame; age is the mean age of the camera frame behind the published pose.

Method	Latency	Rate	1× playback Success (now)	 Lost	 Median (cm)	Age	2× playback Success (now)	 Lost
Static (no motion)	–	–	19%	76%	21.99	–	19%	76%
POGS	526 ms	1.9 Hz	56% ± 19	19%	9.16	802 ms	20%	59%
POGS, 10 iters	225 ms	4.4 Hz	**100%** ± 1	**0%**	1.69	354 ms	25%	48%
POGS + KF + check	540 ms	1.9 Hz	66% ± 2	1%	2.89	824 ms	26%	42%
+ latency comp.	540 ms	1.9 Hz	91% ± 3	**0%**	2.09	824 ms	**34%**	**29%**
POGS + KF + check, 10 iters	253 ms	4.0 Hz	97% ± 3	**0%**	1.86	396 ms	20%	53%
+ latency comp.	253 ms	4.0 Hz	**100%** ± 0	**0%**	**1.49**	396 ms	23%	49%

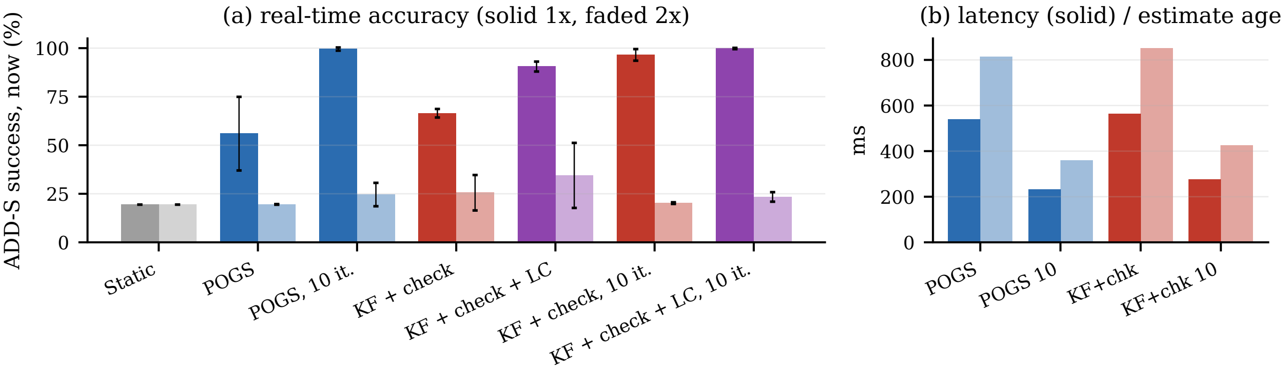
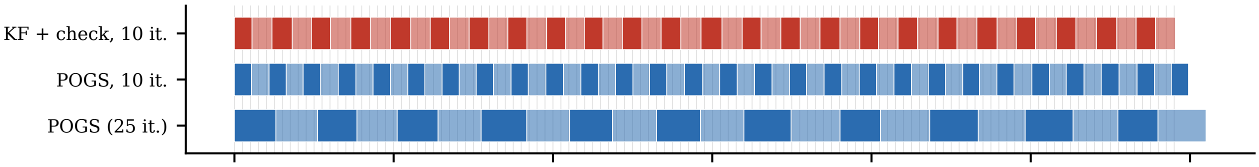


Figure 12: Real-time summary over three seeds. (a) ADD-S success scored at every camera frame (solid 1×, faded 2× playback). (b) Latency per tracked frame (solid) and mean estimate age (faded).

(a) when each tracker is busy (one bar = one tracked camera frame; grid = every 3rd camera frame)



(b) age of the camera frame behind the published pose

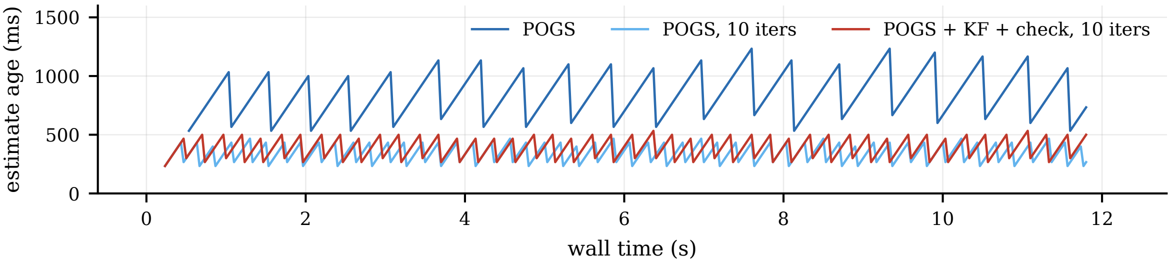


Figure 13: Real-time schedule, base seed 0, 1×. (a) Compute intervals: each bar is one tracked camera frame; every other bar is lighter to show the boundaries. (b) Age of the camera frame behind the published pose: it rises until the next estimate is published, then drops.

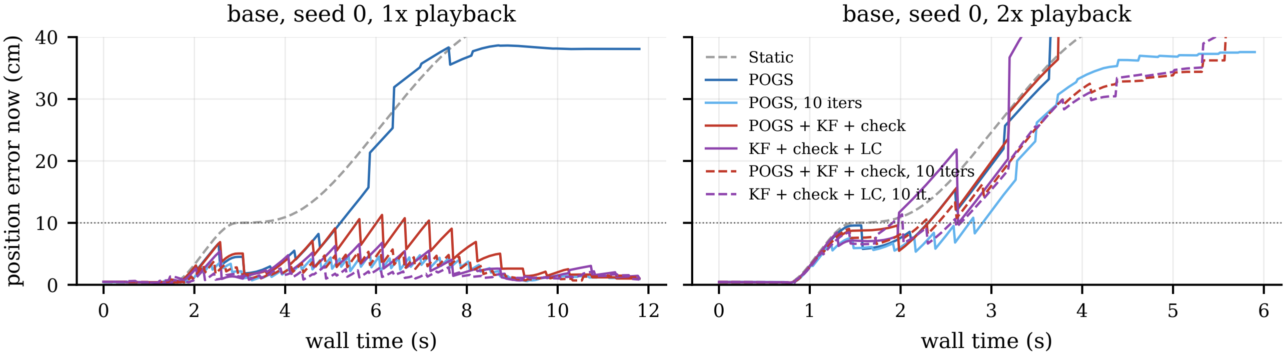
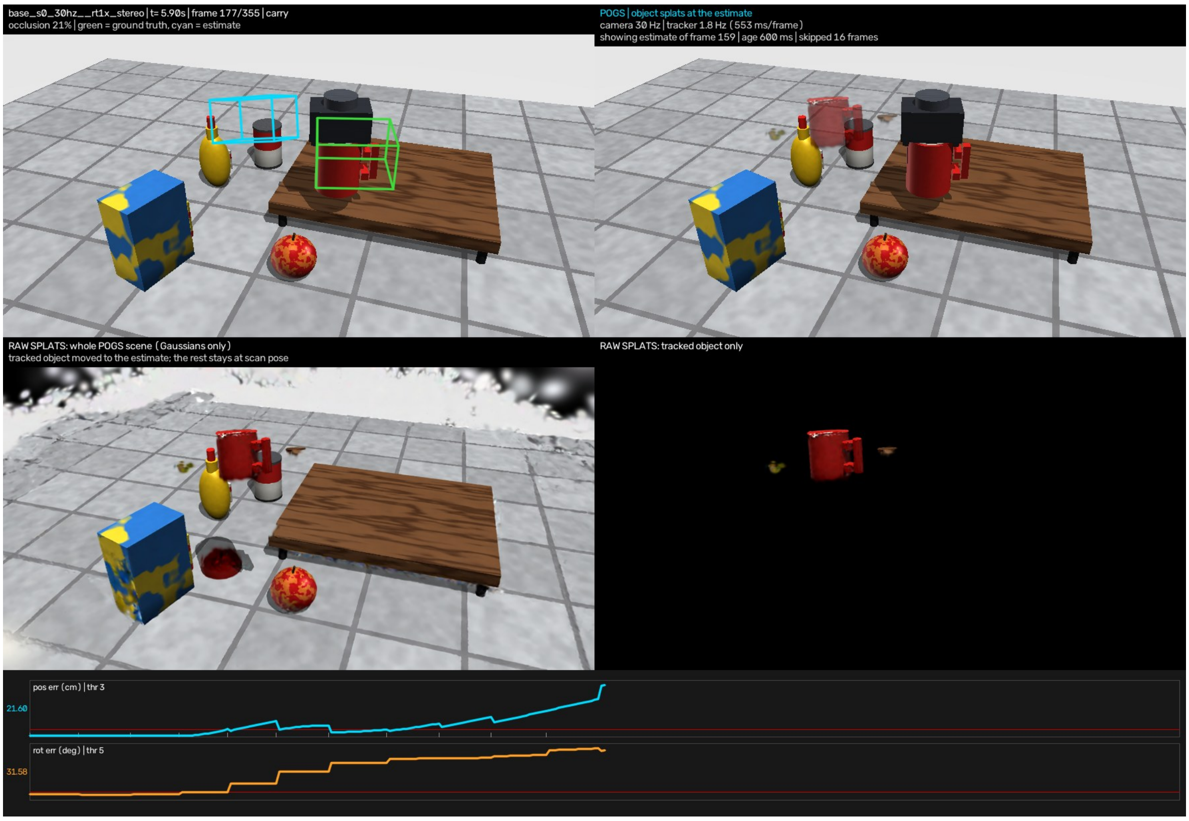
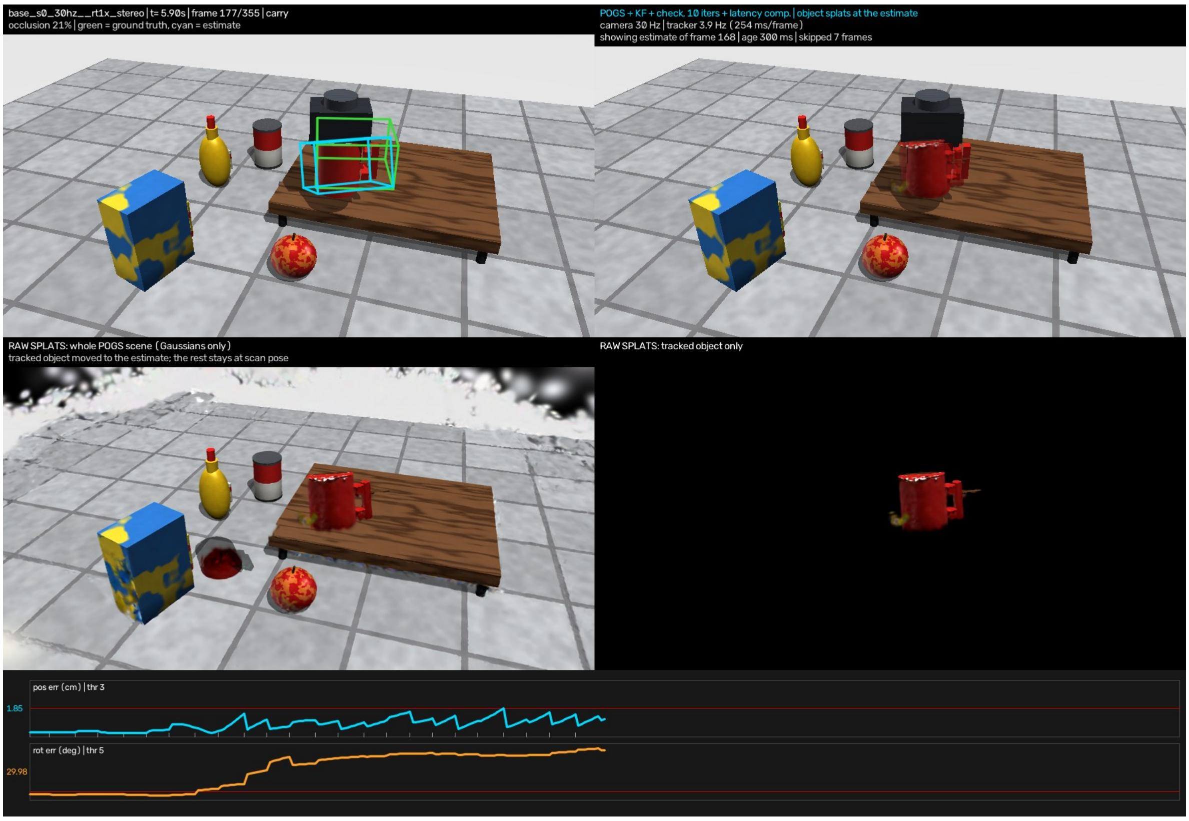


Figure 14: Position error of the published pose at every camera frame, base seed 0. LC = latency compensation. Dashed purple and red: the 10-iteration KF variants.


base_s0_30hz__rt1x_stereo | t= 5.90s | frame 177/355 | carry
occlusion 21% | green = ground truth, cyan = estimate
POGS | object splats at the estimate
camera 30 Hz | tracker 1.8 Hz (553 ms/frame)
showing estimate of frame 159 | age 600 ms | skipped 16 frames
RAW SPLATS: whole POGS scene (Gaussians only)
tracked object moved to the estimate; the rest stays at scan pose
RAW SPLATS: tracked object only
pos err (cm) | thr 3
21.60
rot err (deg) | thr 5
31.58

(a) Pogs (25 iterations, 1.8 Hz): the shown estimate is 600 ms old (16 frames skipped), and the object splats sit on the soup can while the mug is already on the table.


base_s0_30hz__rt1x_stereo | t= 5.90s | frame 177/355 | carry
occlusion 21% | green = ground truth, cyan = estimate
POGS + KF + check, 10 iters + latency comp. | object splats at the estimate
camera 30 Hz | tracker 3.9 Hz (254 ms/frame)
showing estimate of frame 168 | age 300 ms | skipped 7 frames
RAW SPLATS: whole POGS scene (Gaussians only)
tracked object moved to the estimate; the rest stays at scan pose
RAW SPLATS: tracked object only
pos err (cm) | thr 3
1.85
rot err (deg) | thr 5
29.98

(b) Pogs + KF + check, 10 iterations + latency compensation (3.9 Hz): the estimate is 300 ms old and extrapolated to now; the splats follow the mug.

Figure 15: Real-time dashboards, base seed 0, frame 177 of 355 (mid-carry). The panels show the camera image with boxes, the splat overlay, the raw full-scene splats, and the raw object splats, above the error timeline.

7 Discussion

What the benchmark shows about POGS. On slow motion, Pogs is accurate: sub-centimetre median error, robust to depth noise, two objects at once, and a thin bottle. Its limits are those the paper names, and the benchmark now quantifies them. Accuracy collapses once the per-frame motion exceeds what 25 Adam steps can follow (2–4×). A hand covering the object loses it outright. Large rotations that expose unscanned faces are unreliable, with high seed variance. Compute time turns the speed limit into a real-time problem: at 1.9 Hz on a 30 Hz stream, even normal-speed motion is tracked on only about half of the frames.

What the Kalman prior does and does not do. A warm start helps only when the optimiser can verify and refine it. With the loss check and at least 10 iterations, it is the best method overall and solves the 2× and cereal-box failures. Without the check, or with 5 iterations, it is unsafe. The optimiser's output is not independent of the prediction, so a wrong prediction confirms itself, and the filter carries a lost object away at the speed limit. Damping and speed limits bound the damage but do not remove it. The real-time results put the gains in proportion: making the tracker fast (10 iterations) is worth more than the motion prior, and the cheapest single improvement is extrapolating the published pose to the current time.

Limitations of this study.

- The data are synthetic: one scene, one trained model, rigid objects, a static camera, and an analytic renderer. Absolute numbers are not comparable with the paper's physical-robot results, which measured placement error with calipers (2.5–4.1 cm, Table I of [1]) and tool-servo success rates.
- Three seeds per level give large standard deviations on the hard levels; differences within one standard deviation should not be over-read.
- Grouping uses rendered instance masks and geometric selection instead of Detic and an operator's click.
- Real-time timings are for one RTX 4070 and include only tracking compute (DINO features + optimisation). Initialisation is untimed.
- Rotation error is dominated by the gripper-induced tilt, so the paper's 3 cm & 5° rule cannot rank methods here.

Next steps. Occlusion is the clearest open problem: no variant handles a hand over the object. Promising directions include occlusion-aware depth residuals (masking the known gripper, or rejecting depth in front of the rendered object), holding the motion prior's pose instead of optimising while the object is mostly hidden, and gating Kalman updates by NIS. For real time, the largest gains are likely from cheaper per-frame features and asynchronous extrapolation at the camera rate.

8 Conclusion

We re-implemented Pogs from the paper and the authors' release, and built a controlled synthetic benchmark around the limitations the paper reports. On it, we evaluated a Kalman motion prior and a real-time protocol driven by measured latency. Pogs is accurate on slow motion but fails at higher speeds, under hand occlusion and on large rotations. As published, it is too slow for a live 30 Hz camera: right on 56% of frames, with poses 0.8 s old. A loss-checked SE(3) Kalman warm start is the most accurate method offline (75% vs. 71% success; 100% vs. 82% at twice the speed; 86% vs. 57% on the cereal box). Combined with fewer iterations and latency compensation, it tracks the real-time stream on 100% of frames with the lowest error. The prior is unsafe without the loss check or with a very small iteration budget, and hand occlusion remains unsolved.

Reproducibility. All code, data generators, cached runs and videos are in the repository. The full pipeline runs with

```
.venv/bin/python -m scene_generator.preview_scene       # optional scene preview (CPU)
.venv/bin/python -m experiments.v1_base_pogs.run_all    # data, train, group, track, videos
.venv/bin/python -m experiments.v2_kalman_pogs.run_all
.venv/bin/python -m experiments.v3_scene_mustard.run_all
.venv/bin/python -m experiments.realtime.run_all        # 30 Hz data + real-time protocol
.venv/bin/python -m experiments.results_tables          # tables (CPU)
.venv/bin/python report/make_figures.py                 # this report's figures (CPU)
```

The full suite took about 4.3 h of GPU time on one RTX 4070 (training 12 min, v1 69 min, v2 2.8 h, real time 12 min).

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A Full speed × budget table

Table 9: Speed factor at three budgets: ADD-S success ± seed std / lost.

Level	POGS	KF + check	POGS, 10	KF + check, 10	POGS, 5	KF + check, 5
1×	100±0 / 0	100±0 / 0	100±0 / 0	100±0 / 0	86±24 / 11	38±3 / 47
2×	82±31 / 17	100±0 / 0	50±3 / 43	66±30 / 30	23±0 / 56	50±23 / 31
4×	40±0 / 60	47±7 / 36	27±0 / 56	27±0 / 56	24±4 / 60	27±0 / 58
yaw 90, 1×	49±15 / 21	47±7 / 10	52±15 / 34	48±8 / 37	41±4 / 49	37±3 / 39
yaw 90, 2×	47±6 / 37	48±5 / 46	40±7 / 51	46±8 / 36	22±2 / 58	27±9 / 50
yaw 90, 4×	36±4 / 60	40±0 / 53	24±4 / 60	27±0 / 58	22±4 / 64	24±4 / 58

B Software and data layout

- `scene_generator/`: ray-caster, materials, scene, benchmark conditions, trajectory planner, dataset writer, scene preview.
- `common/`: model, trainer, encoders (DINO, CLIP), grouping, tracking (RigidGroupOptimizer), Kalman filter, evaluation (offline and real-time), checkpoint.
- `experiments/`: v1, v2, v3 and real-time drivers; shared pipeline; dashboard videos; result tables.
- `data_results/`: per-experiment summaries, videos, and the shared run cache (`benchmark_runs/<case>/<method>/`: row, telemetry, poses, real-time schedule).
- `report/`: this document and `make_figures.py`.